Database Technologies – Assignment 9 **KD2-Abhishek-96835**

1. Create an index that will enable a user to pull orders for '1990-10-03' out of the Orders table quickly.

```
KD2_Abhishek_96835>CREATE INDEX ord_odate_idx1 ON orders(odate);
Query OK, 0 rows affected, 1 warning (0.12 sec)
Records: 0  Duplicates: 0  Warnings: 1

KD2_Abhishek_96835>SELECT * FROM orders WHERE odate='1990-10-03';
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
+------+---------+------------+------+------+
5 rows in set (0.00 sec)
```

2. If the Orders table has already been created, how can you force the onum field to be unique (assume all current values are unique)?

```
KD2_Abhishek_96835>CREATE UNIQUE INDEX ord_onum_idx ON orders(onum);
Query OK, 0 rows affected (0.05 sec)
Records: 0  Duplicates: 0  Warnings: 0

KD2_Abhishek_96835>INSERT INTO orders VALUES(3006,100.50,'1990-10-05',2008
,1004);
ERROR 1062 (23000): Duplicate entry '3006' for key 'orders.ord_onum_idx'
KD2_Abhishek_96835>
```

3. Create an index that would permit salesperson to retrieve his orders.

```
KD2_Abhishek_96835>CREATE INDEX ord_snum_idx2 ON orders(snum);
Query OK, 0 rows affected, 1 warning (0.05 sec)
Records: 0  Duplicates: 0  Warnings: 1

KD2_Abhishek_96835>SELECT * FROM orders WHERE snum=1007;
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
+------+---------+------------+------+------+
2 rows in set (0.00 sec)

KD2_Abhishek_96835>EXPLAIN ANALYZE SELECT * FROM orders WHERE snum=1007;
+-----------------------------------------------------------------------------------------------------------------------------------+
| EXPLAIN                                                                                                                           |
+-----------------------------------------------------------------------------------------------------------------------------------+
| -> Index lookup on orders using ord_snum_idx (snum=1007)  (cost=0.7 rows=2) (actual time=0.0338..0.0442 rows=2 loops=1)
 |
+-----------------------------------------------------------------------------------------------------------------------------------+
1 row in set (0.01 sec)
```

4. Let us assume that each salesperson is to have only one customer of a given rating, and this is currently the case. Create an index that enforces it.

```
KD2_Abhishek_96835>CREATE UNIQUE INDEX cust_snum_rating_idx ON customers(snum,rating);
ERROR 1062 (23000): Duplicate entry '1001-100' for key 'customers.cust_snum_rating_idx'
KD2_Abhishek_96835>SELECT * FROM customers;
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2001 | Hoffman  | London   |    100 | 1001 |
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2006 | Clemens  | London   |    100 | 1001 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
| 2007 | Pereira  | Rome     |    100 | 1004 |
+------+----------+----------+--------+------+
7 rows in set (0.00 sec)
```

5. Create an index to speed up searching order on a given date by given customer.

```
KD2_Abhishek_96835>CREATE INDEX ord_odate_cnum_idx ON orders(odate,cnum);
Query OK, 0 rows affected (0.04 sec)
Records: 0  Duplicates: 0  Warnings: 0

KD2_Abhishek_96835>SELECT * FROM orders WHERE odate='1990-10-03' AND cnum=20
01;
+------+--------+------------+------+------+
| onum | amt    | odate      | cnum | snum |
+------+--------+------------+------+------+
| 3003 | 767.19 | 1990-10-03 | 2001 | 1001 |
+------+--------+------------+------+------+
1 row in set (0.01 sec)
```